Domain: Digital Marketplace for services (non-retail)

Problem Statement: On-Demand Service Scheduling & Fulfilment System

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1. Preface

1.1 Document Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define and describe the functional and non-functional requirements of the On-Demand Service Scheduling & Fulfilment System, a digital marketplace designed for non-retail services. This document serves as a formal reference that outlines the expected behavior, constraints, and scope of the system to ensure a common understanding among all stakeholders.
The system aims to provide a unified platform that connects service providers with service consumers, enabling seamless service discovery, scheduling, booking, and fulfilment. By digitizing and automating the end-to-end service lifecycle—from service listing and appointment scheduling to service execution and feedback—the platform seeks to reduce manual coordination, improve service accessibility, and enhance operational efficiency.
This SRS is intended to guide system developers, designers, testers, project managers, and stakeholders throughout the software development lifecycle. It acts as a contractual and technical baseline for system implementation, validation, and future enhancements, ensuring that the developed solution aligns with user needs, business objectives, and quality standards.

1.2 Intended Audience

This Software Requirements Specification (SRS) document is intended for the following stakeholders involved in the development, deployment, and use of the On-Demand Service Scheduling & Fulfilment System:

- **Project Stakeholders and Clients**
  To understand the overall system objectives, scope, and key requirements, and to ensure that business goals and user expectations are accurately captured.

- **System Analysts and Software Architects**
  To use the documented requirements as a basis for system modelling, architectural design, and technology selection.
- **Software Developers**
  To gain a detailed understanding of functional behaviours, system constraints, and interfaces required for implementation.
- **Quality Assurance and Testing Teams**
  To derive test cases, validation criteria, and acceptance tests based on the specified functional and non-functional requirements.
- **Project Managers**
  To support project planning, effort estimation, scheduling, and progress tracking throughout the development lifecycle.

2. Introduction

2.1 Purpose

The purpose of this document is to specify the software requirements of the **On-Demand Service Scheduling & Fulfilment System**; a digital marketplace developed for non-retail service offerings. This SRS provides a detailed description of the system's functional and non-functional requirements to ensure a shared understanding among stakeholders and to serve as a baseline for system design, development, testing, and deployment.

2.2 Scope

The On-Demand Service Scheduling & Fulfilment System aims to provide a centralized digital platform that connects service providers with service consumers. The system enables users to discover services, schedule appointments, place service requests, and track service fulfilment in real time. Service providers can list services, manage availability, accept or reject requests, and complete service delivery through the platform.
The system focuses on non-retail services such as professional, technical, and personal services, excluding the sale of physical goods. It aims to improve service accessibility, reduce manual coordination, and streamline the end-to-end service management process. The platform is designed to be scalable, secure, and adaptable to future enhancements such as analytics, dynamic pricing, and third-party integrations.

2.3 Overview

This SRS document presents an overview of the system architecture, user roles, operating environment, and constraints. It outlines both functional requirements—such as service booking, scheduling, and fulfilment—and non-functional requirements, including performance, security, and usability.

Subsequent sections of this document provide a detailed description of the system features, user interactions, external interfaces, and quality attributes. This structure ensures clarity, traceability, and consistency throughout the software development lifecycle.

3. User Requirement Definition

3.1 User Classes

The On-Demand Service Scheduling & Fulfilment System supports multiple user classes, each with specific roles and responsibilities:

- **Customer**
  An end user who browses available services, books instant or scheduled services, manages personal profiles, and provides ratings and reviews after service completion.
- **Service Provider**
  A user who offers services through the platform. Service providers manage their profiles, define service offerings, set availability using a work calendar, and fulfil customer service requests.
- **Manager**
  A supervisory user responsible for overseeing service operations. The manager monitors service completion, manages service providers, handles customer issues, and generates revenue and performance reports.
- **System (Automated Services)**
  The system handles background operations such as service tracking, scheduling conflict detection, notifications, and failure handling to ensure smooth service fulfilment.

3.2 User Services

The system provides the following core services to its users:

- User registration, authentication, and profile management
- Service discovery and on-demand or scheduled booking
- Real-time service tracking and status updates
- Ratings, reviews, and feedback management
- Issue handling, rescheduling, and cancellation support
- Administrative monitoring and reporting

3.3 System Features

The major system features are organized based on user roles, as represented in the use case diagram.

Customer Features

- Browse and select available services
- Book instant or scheduled services
- Manage personal profile information
- Track service status and completion
- Rate and review completed services

Service Provider Features

- Create and manage service provider profile
- Define and update service offerings
- Manage work calendar and availability
- Accept, perform, and complete service requests

Manager Features

- Monitor daily service completion and performance
- Manage registered service providers
- Handle service scheduling failures and rescheduling
- Manage customer-related issues and disputes
- Generate and analyse revenue and operational reports

System Features

- Track service lifecycle from booking to completion
- Handle scheduling conflicts and automated reassignment
- Maintain service records and system logs
- Ensure system reliability and data consistency

4. System Requirements Specification

4.1 Functional Requirements

4.1.1 Customer Functional Requirements

FR-1: The system shall allow a customer to book a service either instantly or for a scheduled date and time.
FR-2: The system shall allow the customer to provide service details including service type and service location.
FR-3: The system shall allow the customer to select from system-generated available service slots.
FR-4: The system shall allow the customer to make payment for a service booking.
FR-5: The system shall confirm the booking after successful payment.
FR-6: The system shall notify the customer about service provider assignment and schedule details.
FR-7: The system shall allow the customer to submit feedback and ratings after service completion.

4.1.2 Service Provider Functional Requirements

FR-8: The system shall allow service providers to manage their profile information.
FR-9: The system shall allow service providers to view and manage their assigned service calendar.
FR-10: The system shall automatically assign service requests to service providers without requiring provider approval.
FR-11: The system shall allow service providers to update service execution status after performing a service.

4.1.3 Fulfillment & Scheduling Functional Requirements

FR-12: The system shall dynamically generate available service slots based on service duration, provider availability, and regional capacity.
FR-13: The system shall validate selected service slots to prevent time overlaps, double bookings, and capacity conflicts.
FR-14: The system shall lock the selected service slot after successful payment to prevent race conditions.
FR-15: The system shall create a service request entity upon booking confirmation containing service, slot, location, and customer details.

4.1.4 Provider Discovery & Assignment Functional Requirements

FR-16: The system shall automatically analyze required skills for each service request.
FR-17: The system shall identify eligible service providers based on skill

match, location proximity, availability, and workload.
FR-18: The system shall rank eligible service providers using system-defined prioritization metrics such as ratings and reliability.
FR-19: The system shall automatically assign the highest-ranked eligible service provider to the service request.
FR-20: The system shall lock the provider assignment and prevent overrides except by authorized administrators.

4.1.5 Delivery & Execution Functional Requirements

FR-21: The system shall transfer execution responsibility to the delivery team after provider assignment.
FR-22: The system shall track service execution states including Assigned, In-Progress, Completed, Failed, or Cancelled.
FR-23: The system shall allow service providers to update service completion status upon execution.

4.1.6 Failure Handling & Recovery Functional Requirements

FR-24: The system shall detect scheduling failures such as provider unavailability, delay, or no-show.
FR-25: The system shall attempt automatic reassignment to an alternate eligible provider within a defined fallback window.
FR-26: The system shall reschedule the service or cancel the booking if reassignment is not possible.
FR-27: The system shall notify the customer about reassignment, rescheduling, or cancellation events.

4.1.7 Manager (Collective Manager) Functional Requirements

FR-28: The system shall track daily service completion metrics for all service providers.
FR-29: The system shall allow the manager to manage service provider skill sets.
FR-30: The system shall generate revenue reports based on completed services.
FR-31: The system shall allow the manager to view and analyze collective revenue and performance reports.

4.2 Non-Functional Requirements

4.2.1 Performance Requirements

NFR-1: The system shall process service booking requests within an acceptable response time.
NFR-2: The system shall complete provider assignment within a short duration after booking confirmation.

4.2.2 Reliability & Availability Requirements

NFR-3: The system shall always be available except during scheduled maintenance periods.
NFR-4: The system shall reliably handle scheduling failures and recovery scenarios without data loss.

4.2.3 Consistency & Concurrency Requirements

NFR-5: The system shall ensure transactional consistency between slot locking, payment processing, and provider assignment.
NFR-6: The system shall prevent race conditions during slot selection and booking confirmation.

4.2.4 Security Requirements

NFR-7: The system shall ensure secure storage and transmission of user data and payment information.
NFR-8: The system shall restrict administrative and managerial functionalities to authorized users only.

4.2.5 Usability Requirements

NFR-9: The system shall provide a user-friendly interface for customers, service providers, and managers.
NFR-10: The system shall ensure that booking and scheduling workflows are simple and easy to understand.

4.2.6 Scalability Requirements

NFR-11: The system shall support growth in the number of users, service providers, and concurrent service requests.

4.2.7 Observability & Maintainability Requirements

NFR-12: The system shall log fulfilment decisions including provider ranking and assignment outcomes.

NFR-13: The system shall be designed to allow easy updates and maintenance without impacting core functionality.

5. System Models

5.1 Use Case Diagram

The use case diagram represents the interaction between different actors and the on-demand service platform. The primary actors involved in the system are **Customer, Service Provider, System Admin, Unit Manager, and Collective Manager**. The functionalities available to each actor are described below.

Customer Use Cases

Use Case	Description
Browse and Discover Services	Allows customers to explore available services, categories, and service providers before making a selection.

Add Services to Cart	Enables customers to select one or more services and add them to a cart before confirming the booking.
Checkout	Allows the customer to finalize the booking after selecting services from the cart and confirming service details.
Rate and Review Service	Enables customers to provide feedback and ratings after service completion to maintain service quality.

Service Provider Use Cases

Use Case	Description
Profile Management	Allows service providers to manage personal information, service details, skills, and required documents.
Join a Unit	Enables service providers to request membership in a specific unit within a collective to receive service assignments.
View Earnings	Allows service providers to track earnings from completed services and view payment summaries.

Manage Work Calendar	Enables service providers to manage their availability by marking time slots as available or unavailable.

System Admin Use Cases

Use Case	Description
Background Data Verification	Enables the admin to verify the authenticity of service provider documents and background information.
Handle Schedule Failure / Reassignment	Allows the admin to reassign services when a provider cancels or becomes unavailable.
View Complaints / User Care	Enables the admin to review complaints from users and provide support or resolutions.
Suspend Accounts	Allows the admin to suspend accounts that violate policies or misuse the platform.

Unit Manager Use Cases

Use Case	Description
Manage Work Calendar	Allows the unit manager to oversee and manage the availability schedules of service providers within the unit.
Manage Service Providers	Enables the unit manager to monitor and coordinate service providers working within the unit.
Access Revenue Reports	Allows the unit manager to access financial reports related to services performed within the unit.

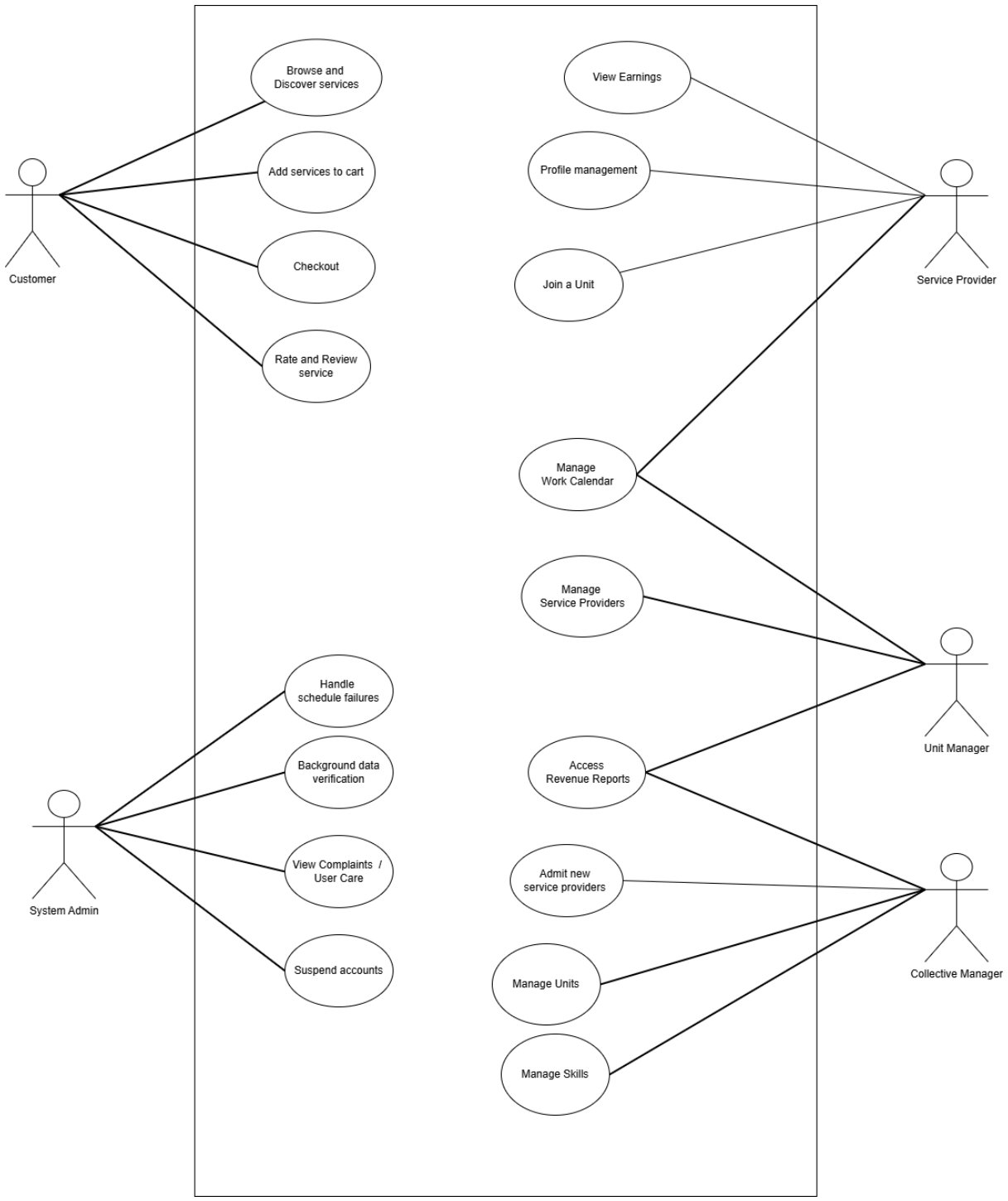
Collective Manager Use Cases

Use Case	Description
Admit New Service Providers	Allows the collective manager to approve and onboard new service providers into the collective.
Manage Units	Enables the collective manager to create, update, and manage units within the collective.

Manage Skillset	Allows the collective manager to maintain service categories and skillsets available on the platform.
Manage Revenue Reports	Enables the collective manager to generate and analyze overall revenue reports across units.

Summary

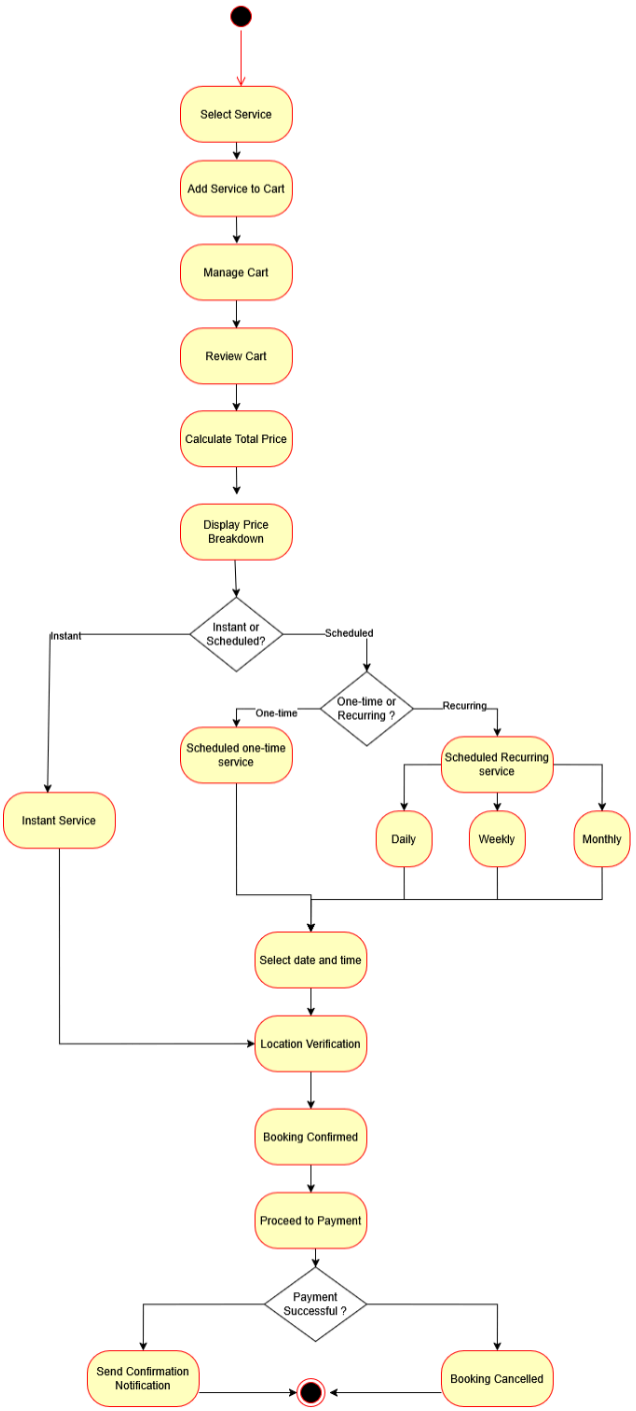
These use cases collectively define the interactions between system actors and the on-demand service platform. The system supports efficient **service discovery, booking, provider onboarding, scheduling management, complaint handling, and revenue monitoring**, ensuring smooth operation of the platform.


Customer
Browse and Discover services
Add services to cart
Checkout
Rate and Review service
Service Provider
View Earnings
Profile management
Join a Unit
Unit Manager
Manage Work Calendar
Manage Service Providers
Access Revenue Reports
Collective Manager
Admit new service providers
Manage Units
Manage Skills
System Admin
Handle schedule failures
Background data verification
View Complaints / User Care
Suspend accounts

5.2 Activity Diagrams

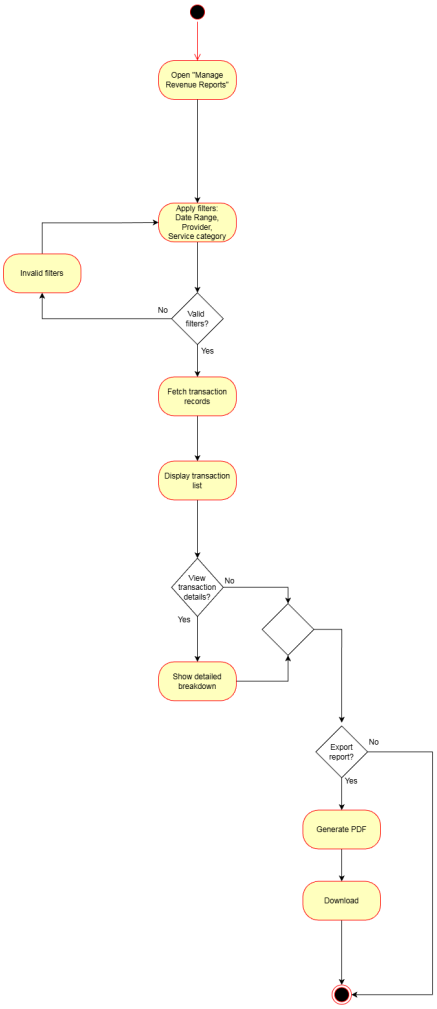
Book instant or scheduled service

Shows how the **Customer** selects a service, chooses instant or scheduled booking, provides details, and completes or cancels the booking based on payment.



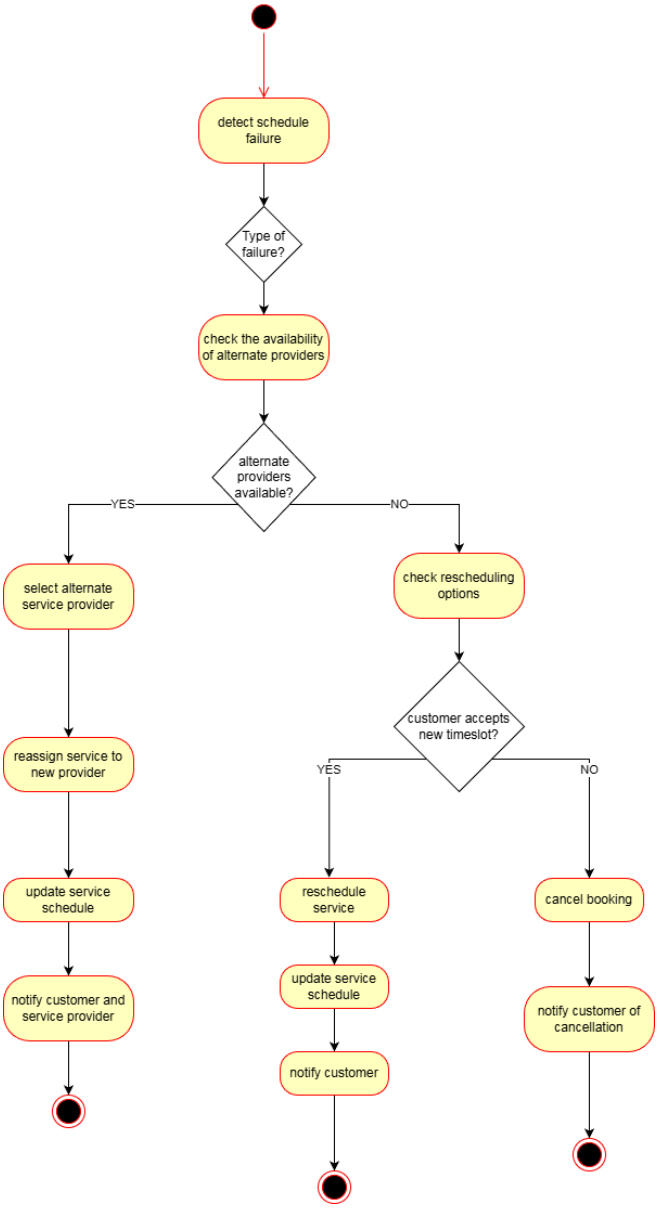
Manage Revenue Report

Shows how the **Manager** views revenue reports generated by the system and optionally exports them.



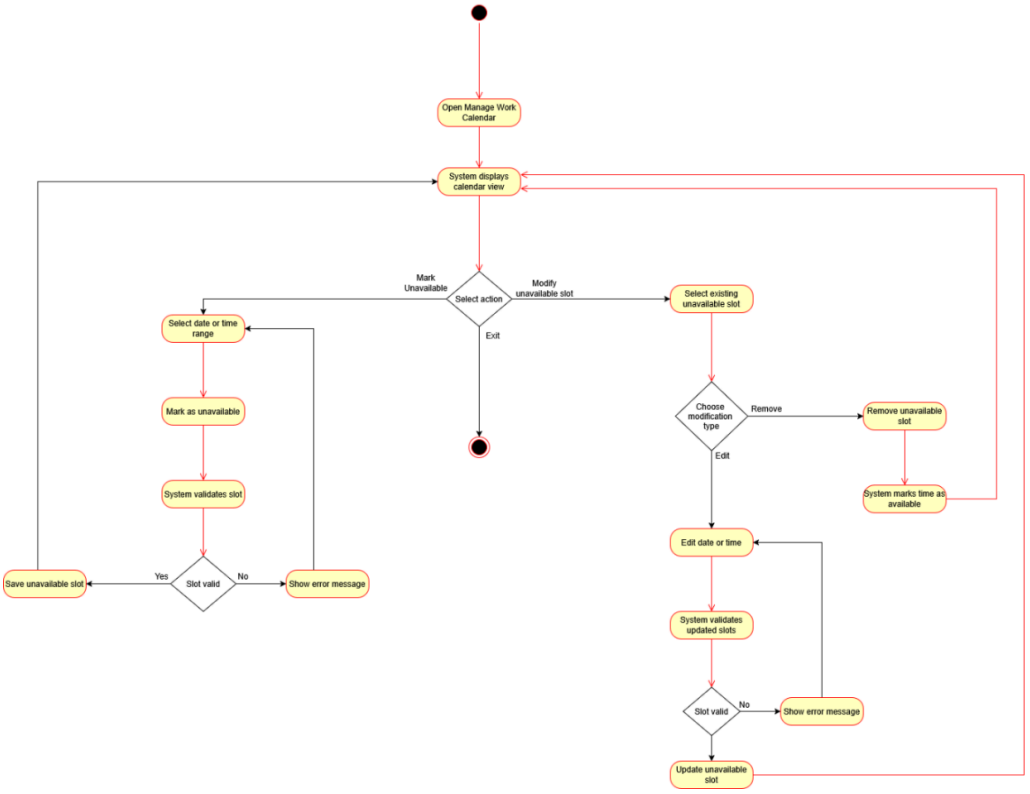
Handle Schedule Failure _ Reassignment

Shows how the **Admin/System** reassigns a service to anot
her provider or reschedules or cancels the booking when a failure occurs.



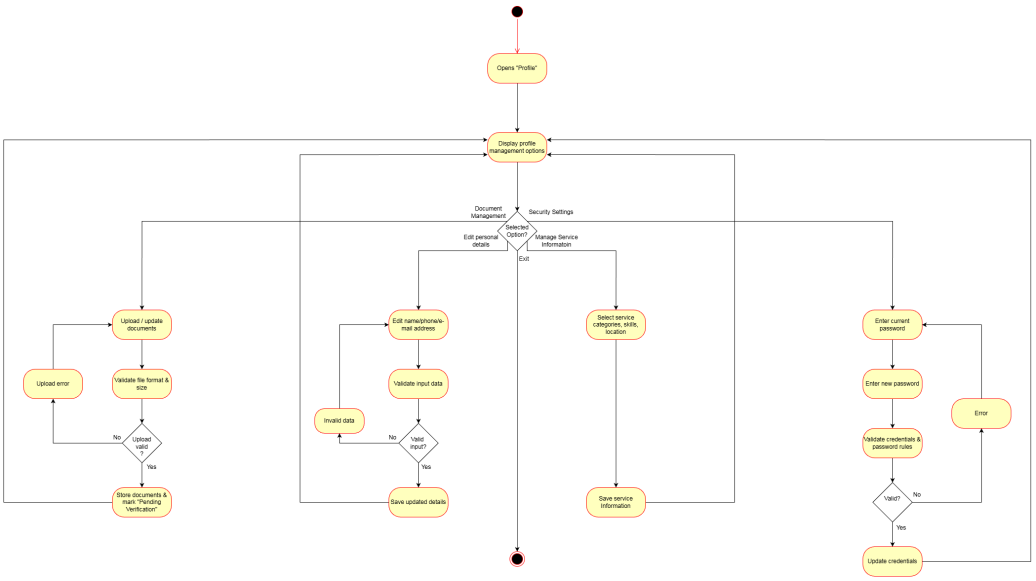
Manage Work Calendar

Shows how the **Service Provider** marks or modifies unavailable time slots to control service availability.



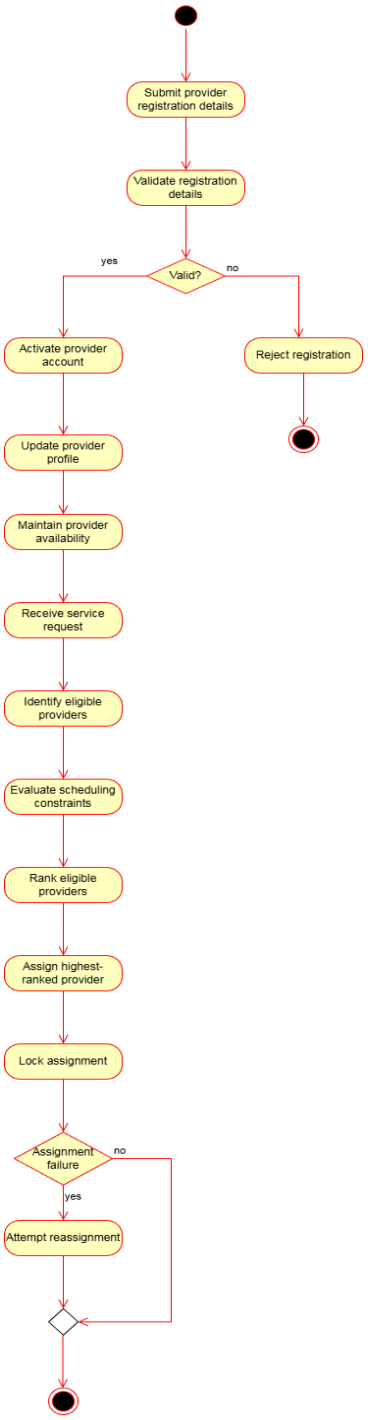
Profile management

Shows how the **Service Provider** updates profile details, service information, documents, and security settings.

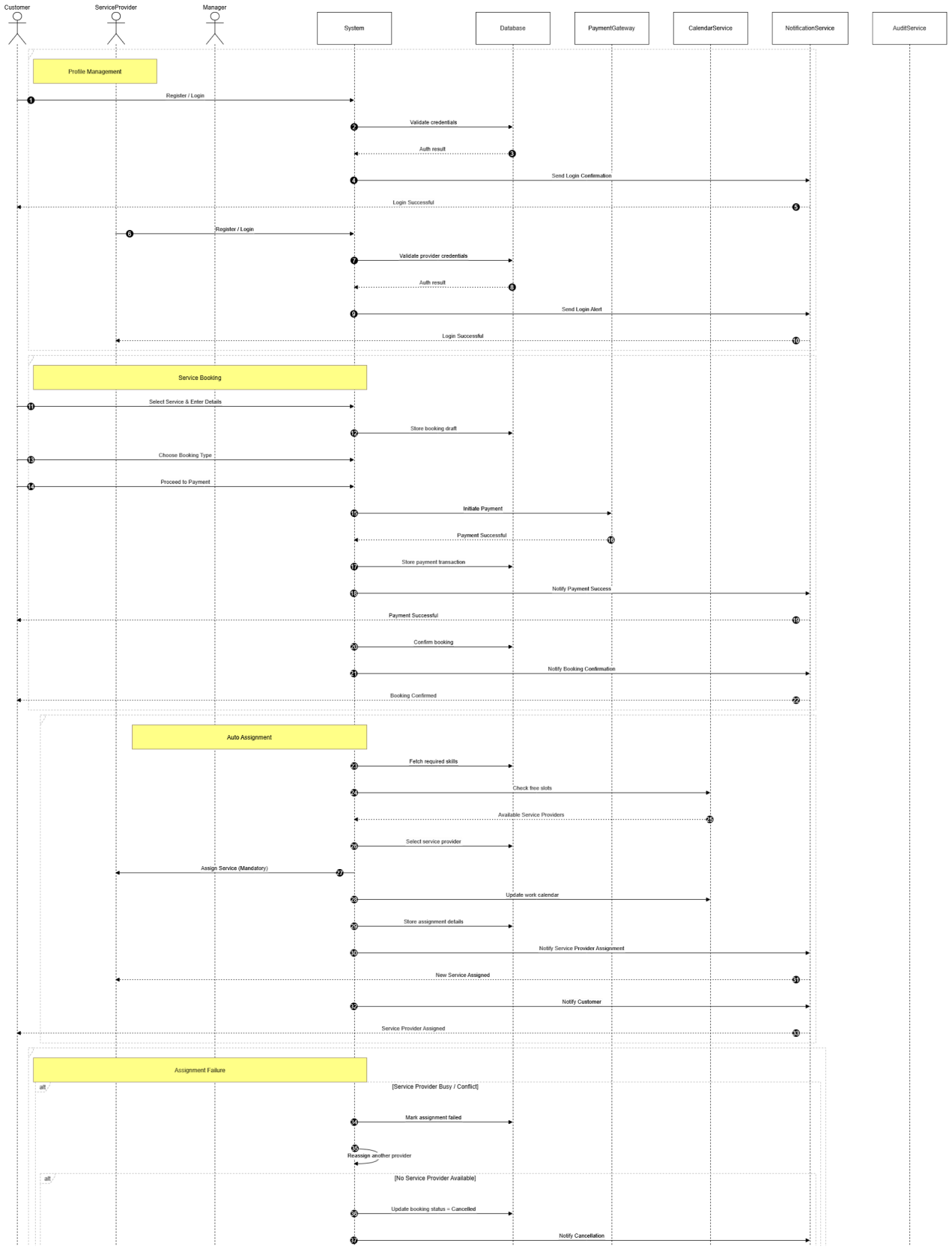


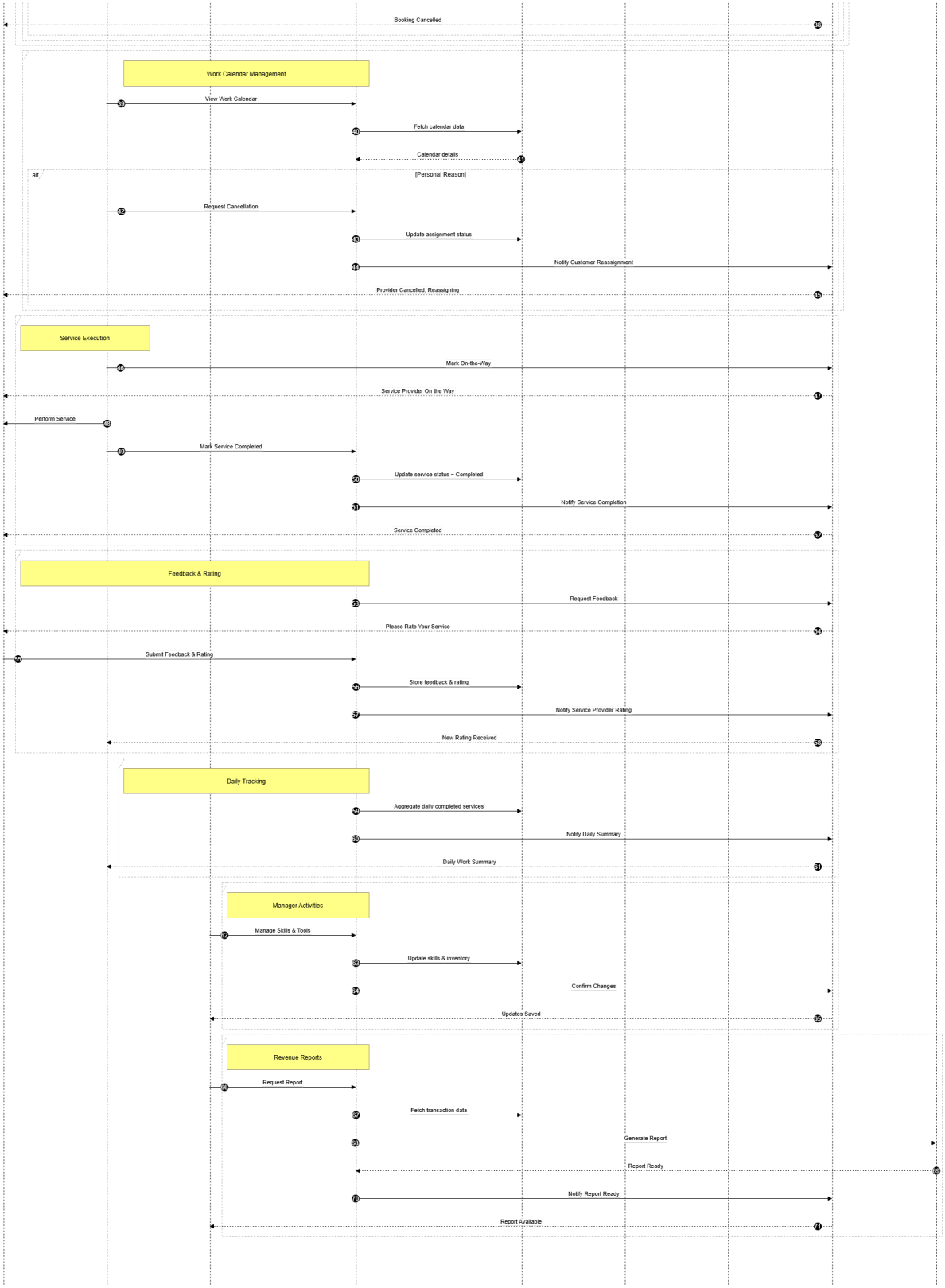
Manage Provider

Shows how the **Admin/System** validates provider registration, activates approved providers, manages availability, and assigns suitable providers to service requests, with reassignment handled in case of failure.



5.3 Sequence Diagram


Customer
ServiceProvider
Manager
System
Database
PaymentGateway
CalendarService
NotificationService
AuditService
Profile Management
1 Register / Login
2 Validate credentials
3 Auth result
4 Send Login Confirmation
5 Login Successful
6 Register / Login
7 Validate provider credentials
8 Auth result
9 Send Login Alert
10 Login Successful
Service Booking
11 Select Service & Enter Details
12 Store booking draft
13 Choose Booking Type
14 Proceed to Payment
15 Initiate Payment
16 Payment Successful
17 Store payment transaction
18 Notify Payment Success
19 Payment Successful
20 Confirm booking
21 Notify Booking Confirmation
22 Booking Confirmed
Auto Assignment
23 Fetch required skills
24 Check free slots
25 Available Service Providers
26 Select service provider
27 Assign Service (Mandatory)
28 Update work calendar
29 Store assignment details
30 Notify Service Provider Assignment
31 New Service Assigned
32 Notify Customer
33 Service Provider Assigned
Assignment Failure
alt
[Service Provider Busy / Conflict]
34 Mark assignment failed
35 Reassign another provider
alt
[No Service Provider Available]
36 Update booking status = Cancelled
37 Notify Cancellation


Booking Cancelled
Work Calendar Management
View Work Calendar
Fetch calendar data
Calendar details
alt
[Personal Reason]
Request Cancellation
Update assignment status
Notify Customer Reassignment
Provider Cancelled, Reassigning
Service Execution
Mark On-the-Way
Service Provider On the Way
Perform Service
Mark Service Completed
Update service status = Completed
Notify Service Completion
Service Completed
Feedback & Rating
Request Feedback
Please Rate Your Service
Submit Feedback & Rating
Store feedback & rating
Notify Service Provider Rating
New Rating Received
Daily Tracking
Aggregate daily completed services
Notify Daily Summary
Daily Work Summary
Manager Activities
Manage Skills & Tools
Update skills & inventory
Confirm Changes
Updates Saved
Revenue Reports
Request Report
Fetch transaction data
Generate Report
Report Ready
Notify Report Ready
Report Available